

MUHAMAD AKMAL HARITH BIN MOHD BAKHTIAR

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PROFESSIONAL SUMMARY

Final-year Electronic Engineering student at UiTM (CGPA 3.51, multiple Dean's List awards) with hands-on experience across machine learning, RF/microwave design, and embedded IoT systems. Has built end-to-end AI systems (computer vision and sensor-based classification with TensorFlow and scikit-learn) and designed sub-6 GHz 5G RF circuits — a low-noise amplifier and a bandpass filter — in Keysight Genesys, alongside hardware testing and troubleshooting from industrial internships. Seeking a Graduate Engineer role in AI/ML, semiconductor, or RF/electronics.

EDUCATION

Bachelor of Electronic Engineering (Hons) Universiti Teknologi MARA (UiTM) CGPA 3.51 • Dean's List Award (Semesters 3, 6 & 7) • Expected graduation 2026	2023 – 2026 Shah Alam, Selangor
Diploma in Electrical (Electronic) Engineering (Hons) Universiti Teknologi MARA (UiTM)	2020 – 2023 Dungun, Terengganu
Technical Institute of Kuala Lumpur (TIKL) Secondary / Technical Education	2018 – 2019 Cheras, Kuala Lumpur

TECHNICAL SKILLS

Machine Learning & AI: TensorFlow, Keras/CNN, scikit-learn, computer vision & image classification, feature engineering (MFCC, DWT, Relief-F), model evaluation (cross-validation, precision/recall/F1)

RF & Microwave: S-parameter analysis, Smith-chart impedance matching, microstrip / transmission-line design, LNA and bandpass-filter design, stability (K-factor), VSWR, return/insertion loss, noise figure

Programming & Data: Python, C/C++, NumPy, Pandas, SciPy, SQLite, data analysis

EDA & Simulation: Keysight Genesys, Multisim, LTspice, Proteus, AutoCAD, Cisco Packet Tracer

Embedded & IoT: ESP32, ESP32-CAM, LoRa, sensor integration, Flask, REST/JSON APIs, Blynk

Test & Measurement: Oscilloscope, signal generator, multimeter, hardware diagnostics, functional testing, troubleshooting

PROJECTS

FloodGuard AI — Intelligent Flood Monitoring & Early-Warning System Technologies: Python, scikit-learn, Flask, SQLite, ESP32, LoRa, HTML/CSS - Built an IoT flood monitoring and early-warning system using ultrasonic and rain sensors on ESP32 nodes with 923 MHz LoRa telemetry; secured official Copyright Registration for the system. - Trained and evaluated a Random Forest classifier (scikit-learn) to classify flood risk into Low / Medium / High from water-distance, rainfall, and rate-of-change features. - Deployed a Flask server (HTTP/JSON) with an SQLite-backed web dashboard for live water level, rainfall, and risk status; automated high-risk LED/siren alerts via relay control. - Validated the wireless link outdoors using RSSI, SNR, response-delay, and packet-reception metrics on a solar-powered field prototype.	<i>Final Year Project</i>
Doctor Plants — AI Tomato Leaf Disease Detection Technologies: ESP32-CAM, TensorFlow, CNN, Python, Flask, Blynk, JSON - Trained a TensorFlow CNN to classify tomato-leaf diseases (early blight, late blight, curl virus, bacterial spot) from live ESP32-CAM image captures. - Served predictions from a Flask backend (HTTPS/JSON) to a web dashboard and Blynk mobile app, returning disease label, confidence score, image, and treatment recommendation. - Added non-leaf image rejection, a searchable prediction history with CSV export, and light/dark UI modes.	<i>Computer Vision</i>
3.6 GHz Low-Noise Amplifier for 5G Receiver Applications Technologies: Keysight Genesys, Qorvo QPL9057, S-Parameters, Smith Chart, Microstrip - Designed and simulated a 3.6 GHz LNA (Qorvo QPL9057) for sub-6 GHz 5G receivers, analysing S-parameters for gain, matching, and stability. - Designed input/output impedance-matching networks via Smith chart and converted lumped elements into distributed microstrip transmission lines.	<i>RF Design</i>

- Achieved **VSWR ≈ 1.875 (target < 2)** and improved the design to **K ≈ 1.001 (unconditional stability)** at 3.6 GHz; produced a fabrication-ready layout.

3.5 GHz Interdigital Chebyshev Bandpass Filter for 5G

RF Filter Design

Technologies: Keysight Genesys, Chebyshev, Coupled-Line Resonators, FR4 PCB

- Designed a ninth-order 0.2 dB Chebyshev interdigital bandpass filter at 3.5 GHz using nine quarter-wave coupled-line microstrip resonators on FR4 (ϵ_r 4.5, 1.6 mm, $\tan\delta$ 0.017).
- Derived coupling coefficients, even/odd-mode impedances, and resonator dimensions; simulated and optimised S11/S21, bandwidth, and stopband attenuation in Keysight Genesys.

Acoustic Vehicle Fault Detection — Research & Methodology Study

ECM613

Technologies: MFCC, DWT, Relief-F, Extreme Learning Machine

- Surveyed Scopus-indexed literature and documented a methodology for classifying vehicle faults from engine-audio signals using MFCC/DWT features, Relief-F feature selection, and an Extreme Learning Machine classifier.

ASEM Semiconductor Talk Program

Industry Exposure (2025)

Technologies: Semiconductor Fabrication, Testing, Packaging, Quality Control

- Gained practical exposure to semiconductor fabrication, testing, packaging, and industrial quality-control practices, and to overall production-flow and process-engineering fundamentals.

WORK EXPERIENCE

Technician (Internship)

Aug 2025 – Oct 2025

Rentwise Sdn. Bhd.

Shah Alam, Selangor

- Performed hardware diagnostics, functional testing, refurbishment, and troubleshooting on IT assets in a Linux environment, supporting lifecycle-management workflows.
- Worked with the technical team to restore, validate, and maintain refurbished systems; supported sustainability-focused lifecycle initiatives.

Solar Engineering Intern

Mar 2023 – Jul 2023

Sunsource Sdn. Bhd.

Petaling Jaya, Selangor

- Designed rooftop solar-panel layouts in AutoCAD and HelioScope and ran energy-efficiency analysis to optimise panel placement and system performance.
- Collaborated with senior engineers on renewable-energy feasibility studies.

AWARDS & CERTIFICATIONS

- Silver Medal — International Research and Information Science Expo
- Copyright Registration — Flood Prediction and Monitoring System
- Dean's List Award — Semesters 3, 6 & 7
- Omada SDN Certified Training Completion • CIDB Green Card Holder

REFERENCES

Dr. Siti Musliha Aishah Musa — Senior Engineer, ams OSRAM Malaysia • 018-777 9971 • siti.musa@ams-osram.com

Dr. Lucyantie Mazalan — Academic Advisor (Bachelor), UiTM • 012-608 1841 • lucyantie@uitm.edu.my

PM Ir. Ts. Dr. Suzi Seroja Sarnin — Final Year Project Supervisor, UiTM • 012-255 2924 • suzis045@uitm.edu.my